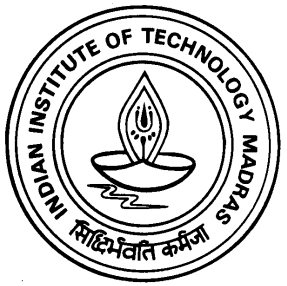
EE2001 - Digital systems lab

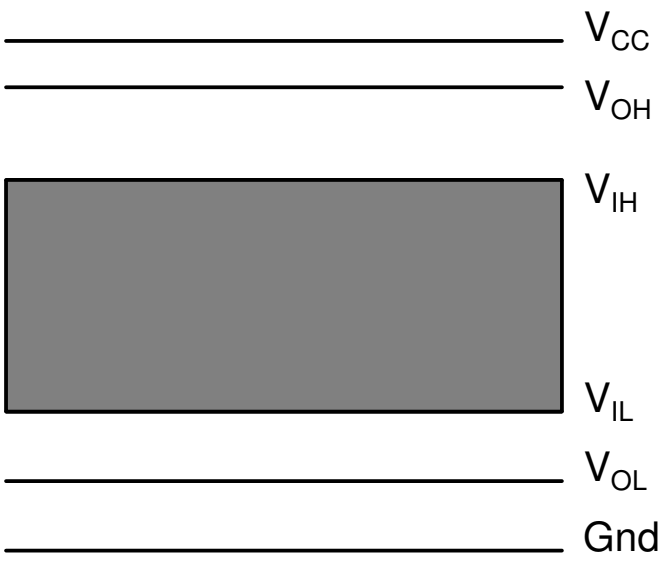
Vinita Vasudevan



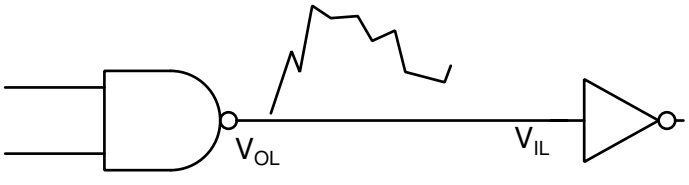
Practical aspects

- What is the range of voltages representing a logic high and low?
- What is the maximum number of gates that can be connected to the output of a gate? (Fanout)
- Propogation delay
- Power consumed

Range of voltages corresponding to logic high and low levels - different for different logic families. You will be using TTL or LSTTL gates.

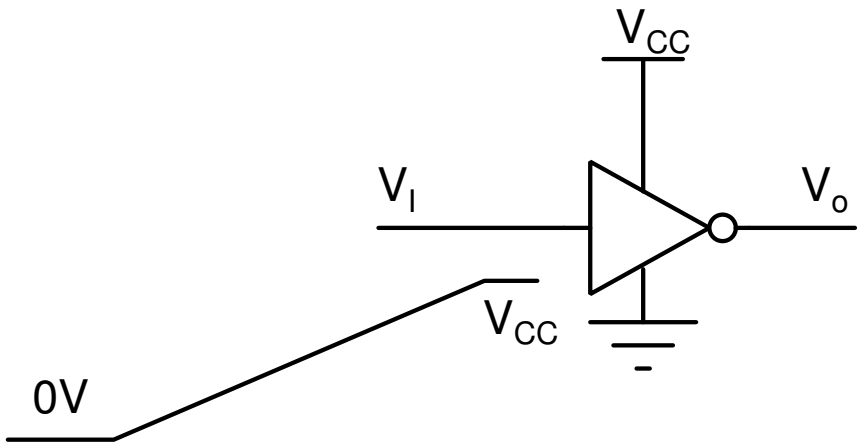


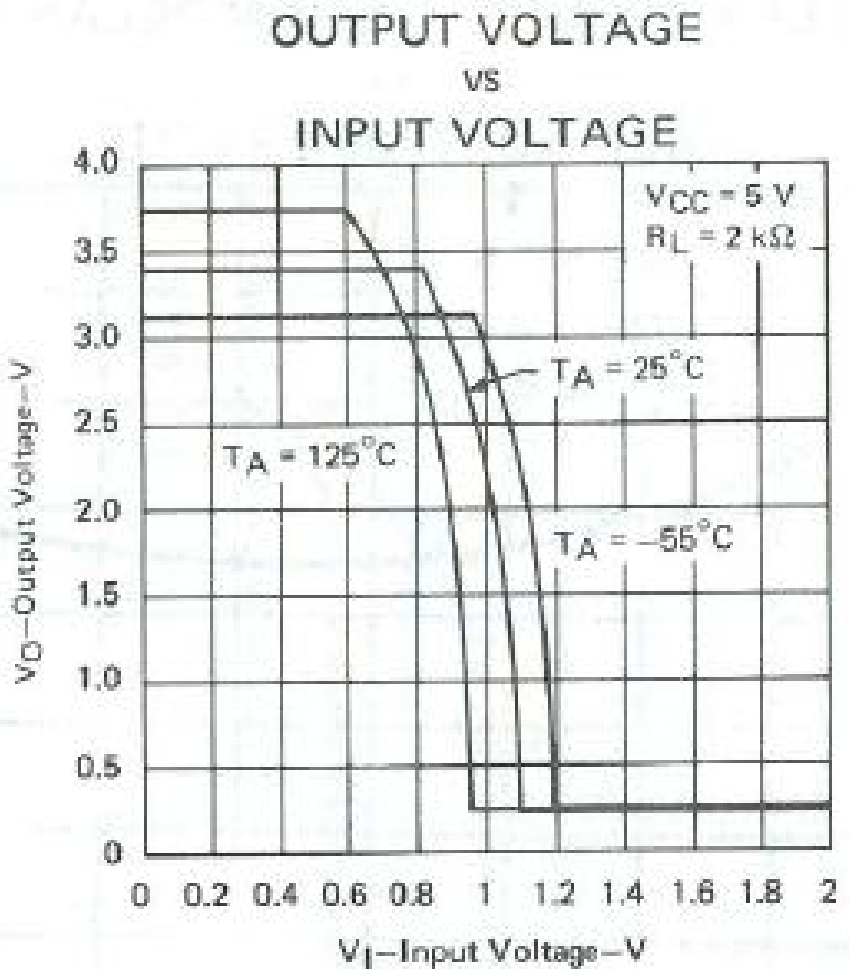
- Typical output high and low voltages (V_{OH} and V_{OL})
- Lowest output voltage that will be regarded as a logic 1 at the input. (V_{IH})
- Highest output voltage that will be regarded as a logic 0 (V_{IL})



Noise margins:
$NM_H = V_{OH} - V_{IH}$,
$NM_L = V_{IL} - V_{OL}$

Voltage transfer characteristics

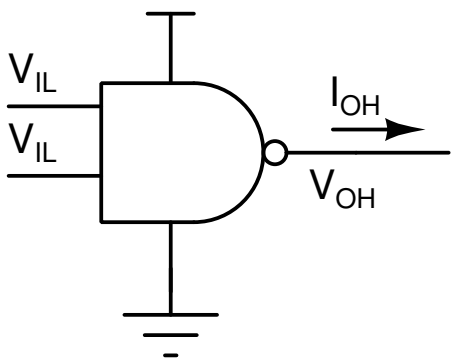


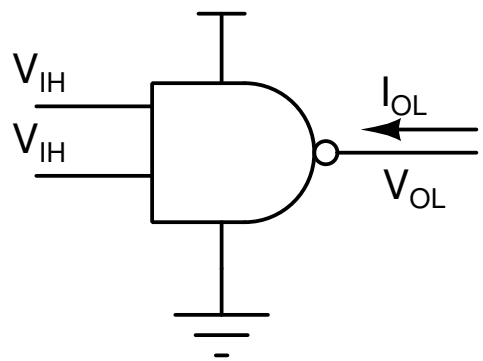


Static characteristics - input voltage is changed slowly.

Practical gates - 74xx LSTTL

	MIN	NOM	MAX	Units
Supply Voltage V_{CC}	4.75	5	5.25	V
High level output current I_{OH}			-400	μA
Low level output current, I_{OL}			8	mA
Operating free air temperature	0		70	$^{\circ}C$





	MIN	NOM	MAX	Units
High level input voltage V_{IH}	2			V
Low level input voltage V_{IL}			0.8	V
High level output voltage V_{OH}	2.7	3.4		V
Low level output voltage V_{OL}		0.25	0.5	V
High level input current, I_{IH}			20	μA
Low level input current, I_{IL}			-0.4	mA

Output characteristics

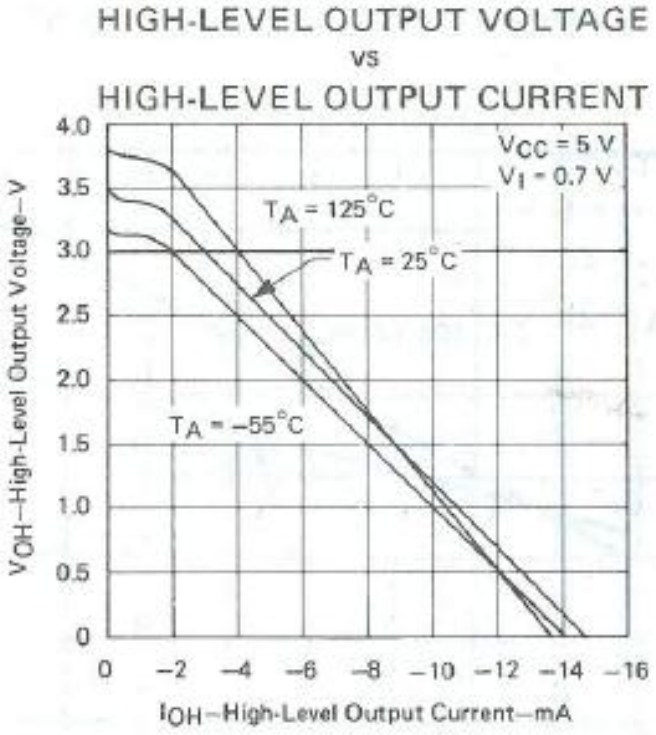


FIGURE D2

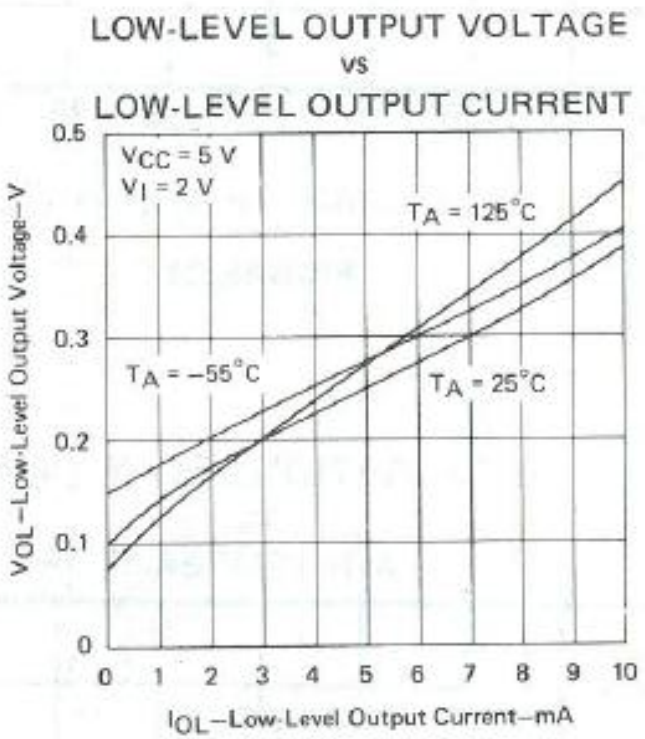
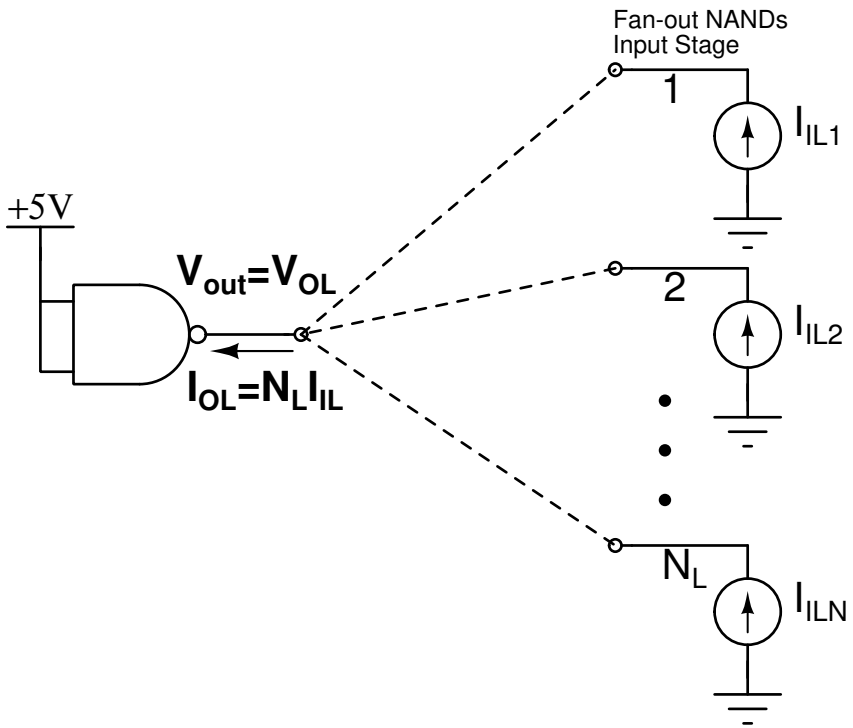
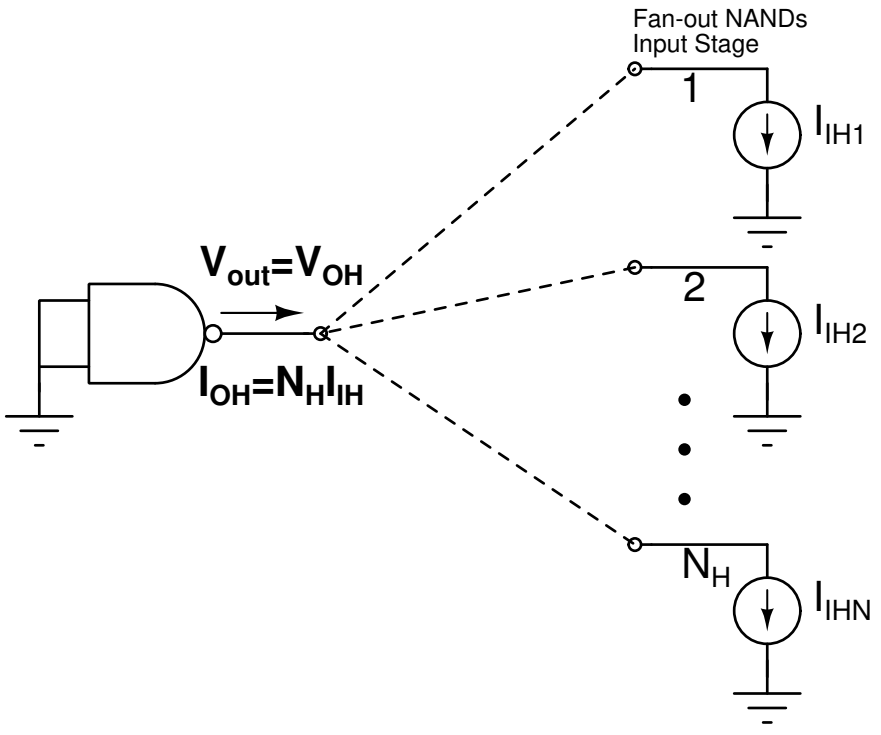


FIGURE D3

Maximum number of gates that can be connected when output voltage is low



Maximum number of gates that can be connected when output voltage is high



Fan-out of the gate is the minimum of $(N_H,\ N_L)$.

Practical Gates: Data sheets

recommended operating conditions

	54 FAMILY / 74 FAMILY	SERIES 54 / SERIES 74 '00, '04, '10, '20, '30			SERIES 54H / SERIES 74H 'H00, 'H04, 'H10, 'H20, 'H30			SERIES 54L / SERIES 74L 'L00, 'L04, 'L10, 'L20, 'L30			SERIES 54LS / SERIES 74LS 'LS00, 'LS04, 'LS10, 'LS20, 'LS30			SERIES 54S / SERIES 74S 'S00, 'S04, 'S10, 'S20, 'S30, 'S133			UNIT
		MIN	NOM	MAX	MIN	NOM	MAX	MIN	NOM	MAX	MIN	NOM	MAX	MIN	NOM	MAX	
Supply voltage, V_{CC}	54 Family	4.5	5	5.5	4.5	5	5.5	4.5	5	5.5	4.5	5	5.5	4.5	5	5.5	V
	74 Family	4.75	5	5.25	4.75	5	5.25	4.75	5	5.25	4.75	5	5.25	4.75	5	5.25	
High-level output current, I_{OH}	54 Family			−400			−500			−100			−400			−1000	μA
	74 Family			−400			−500			−200			−400			−1000	
Low-level output current, I_{OL}	54 Family			16			20			2			4			20	mA
	74 Family			16			20			3.6			8			20	
Operating free-air temperature, T_A	54 Family	−55		125	−55		125	−55		125	−55		125	−55		125	°C
	74 Family	0		70	0		70	0		70	0		70	0		70	

Practical gates: Data sheets

electrical characteristics over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST FIGURE	TEST CONDITIONS†			SERIES 54 SERIES 74 '00, '04, '10, '20, '30			SERIES 54H SERIES 74H 'H00, 'H04, 'H10, 'H20, 'H30			SERIES 54L SERIES 74L 'L00, 'L04, 'L10, 'L20, 'L30			SERIES 54LS SERIES 74LS 'LS00, 'LS04, 'LS10, 'LS20, 'LS30			SERIES 54S SERIES 74S 'S00, 'S04, 'S10, 'S20, 'S30, 'S133			UNIT
						MIN	TYP‡	MAX	MIN	TYP‡	MAX	MIN	TYP‡	MAX	MIN	TYP‡	MAX	MIN	TYP‡	MAX	
V_{IH}	High-level input voltage	1, 2				2			2			2			2			2			V
V_{IL}	Low-level input voltage	1, 2			54 Family			0.8			0.8			0.7			0.7			0.8	V
					74 Family			0.8			0.8			0.7			0.8			0.8	
V_{IK}	Input clamp voltage	3	V_{CC} = MIN, I_I = §					−1.5			−1.5						−1.5			−1.2	V
V_{OH}	High-level output voltage	1	V_{CC} = MIN, V_{IL} = V_{IL} max, I_{OH} = MAX		54 Family	2.4	3.4		2.4	3.5		2.4	3.3		2.5	3.4		2.5	3.4		V
					74 Family	2.4	3.4		2.4	3.5		2.4	3.2		2.7	3.4		2.7	3.4		
V_{OL}	Low-level output voltage	2	V_{CC} = MIN, V_{IH} = 2 V	I_{OL} = MAX	54 Family		0.2	0.4		0.2	0.4		0.15	0.3		0.25	0.4			0.5	V
					74 Family		0.2	0.4		0.2	0.4		0.2	0.4		0.25	0.5			0.5	
				I_{OL} = 4 mA	Series 74LS												0.4				
I_I	Input current at maximum input voltage	4	V_{CC} = MAX		V_I = 5.5 V			1			1			0.1						1	mA
					V_I = 7 V												0.1				
I_{IH}	High-level input current	4	V_{CC} = MAX		V_{IH} = 2.4 V			40			50			10							μA
					V_{IH} = 2.7 V												20			50	
I_{IL}	Low-level input current	5	V_{CC} = MAX		V_{IL} = 0.3 V								−0.18								mA
					V_{IL} = 0.4 V			−1.6			−2						−0.4				
					V_{IL} = 0.5 V															−2	
I_{OS}	Short-circuit output current♦	6	V_{CC} = MAX		54 Family	−20		−55	−40		−100	−3		−15	−20		−100	−40		−100	mA
					74 Family	−18		−55	−40		−100	−3		−15	−20		−100	−40		−100	
I_{CC}	Supply current	7	V_{CC} = MAX			See table on next page															mA